



Bloom

A cycle-aware, evidence-based health coach

All my health data in one place · an AI coach that knows my data, my cycle and the research

Final project · LLM Engineering · Demo Day, 25 September 2026 · Damilya

5 apps, 0 answers

My health data is scattered, and the advice I get is generic and male-default



Withings

weight, body fat,
muscle



Kinetix · Tanita

gym sessions, 1RM
tests, scans



Apple Watch

workouts, cycle, sleep,
HR



Foodvisor

calories & macros (no
API)



Weather

can I run outside?

None of them talk to each other

Weight trends live in one app, training in two others, food in a fourth. No app sees the menstrual cycle next to training and nutrition, so no one can say whether a bad week is the cycle, sleep or under-eating.

Advice ignores women and PCOS

“Train fasted in the morning to burn fat”, “sync your training to your cycle”: popular claims, often from male-only studies or weak evidence. With PCOS, wrong advice (under-fuelling, crash diets) has real costs.

Who it's for

Built for one real user first: me

"A woman with PCOS who strength-trains and runs in Brussels, and wants advice that fits her body, her data and the evidence."

1

Today

Screenshots, spreadsheets and 5 app dashboards; mental math to connect them.

2

Pain

Advice that isn't evidence-graded, ignores the cycle, and never says "see a doctor".

3

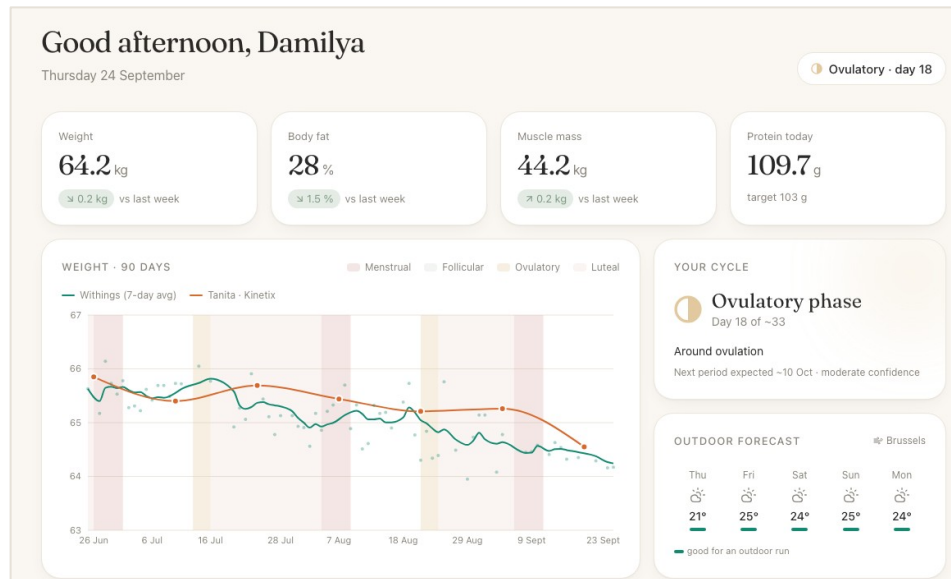
Bloom

One timeline + a coach that cites research, uses my numbers and escalates red flags.

The solution: one calm dashboard + a coach

Next.js app · three personas: Fitness Coach, Nutritionist, Integrative Medicine

- One timeline**
Withings, Tanita, Apple Watch, Kinetix and Foodvisor on one chart, shaded by cycle phase
- Cited answers**
every research claim links to the passage, page and DOI it came from
- Safety first**
red-flag symptoms → doctor referral, never coaching
- Goals you approve**
the coach proposes, you approve or edit before anything is saved



Screenshot: demo data

Demo

Live: bloom-production-4f69.up.railway.app (demo data, password on request)

Should I avoid fasted morning workouts with PCOS?

1

You do not need to strictly avoid fasted morning workouts with PCOS, but there are some important considerations. Evidence does not show that fasted exercise is harmful for women with PCOS, especially for low-intensity sessions, but it is not proven to be superior for fat loss or metabolic health either (Limited) ². For higher-intensity or longer sessions, having a small protein/carb snack beforehand is a reasonable, low-risk default, as women are at higher risk of low energy availability, which can negatively impact menstrual health and performance (Moderate) ⁴.

Why:

- The research on fasted vs. fed exercise is mixed and mostly short-term, with many studies in men or mixed groups, so we cannot make strong, PCOS-specific recommendations (Limited) ².
- For women, especially those with PCOS who may already be at risk for metabolic and reproductive disruptions, maintaining adequate energy intake is important to avoid low energy availability, which can impair menstrual function and overall health (Moderate) ⁴.
- International guidelines for PCOS management recommend regular physical activity as first-line therapy, but do not specify fasted or fed states (Strong) ³.

Next steps:

- If you feel well and enjoy fasted, easy morning workouts, you can continue, but monitor for signs of fatigue, missed periods, or low energy ^{YOUR DATA}.
- For high-intensity or long sessions, try a small snack (e.g., yogurt, banana, or toast) beforehand.
- Track your energy, mood, and cycle patterns to see what works best for you.
- If you notice any cycle disruptions or persistent fatigue, consult your healthcare provider.

This is educational and not a substitute for medical advice.
Educational information, not a diagnosis — for medical decisions, talk to your doctor.

SOURCES

- ² Frampton et al., 2022 - randomized crossover trial
- ³ Patten et al., 2020 - systematic review & meta-analysis
- ⁴ Sims et al., 2023 (ISSN) - position stand

1 • Cited, evidence-graded answer

“Should I avoid fasted morning workouts with PCOS?”

PROPOSED GOAL - NEEDS YOUR APPROVAL

Body fat to 26.5% by 15 December

Body fat: 26.5% · by 2026-12-15

- ✓ 3 resistance training sessions per week (focus on progressive overload)
- ✓ 2 cardio sessions (30–40 min, moderate intensity)
- ✓ Aim for 1.6–1.8g protein/kg body weight daily
- ✓ Prioritize 7.5+ hours sleep per night
- ✓ Track body fat weekly, adjust nutrition if plateau >2 weeks
- ✓ Increase steps to 10,500/day average

You've reduced body fat by 1.5% in the past week, but a safe, sustainable pace is 0.5–1% per month. Reaching 26.5% by mid-December (about 1.5% drop over 10 weeks) is realistic and supports muscle retention, especially with your current training and nutrition habits.

☒ Approve ☐ Edit ☐ Not now

2 • Goal proposal, human approval

“Set me a realistic body-fat goal for December”

I fainted after my workout this morning. Can I still go to the gym tonight?

C

PLEASE CHECK WITH A DOCTOR

I want to pause the coaching here. What you describe — **fainted after workout** — is something a doctor should look at, rather than something I should coach you through.

- If symptoms are happening now or are severe, call 112 (EU emergency number) or go to the nearest emergency department.
- Book an appointment with your GP (huisarts / médecin généraliste) or your gynaecologist/endocrinologist and mention your PCOS.
- Bring your recent data: I can export your cycle history, weight trend and training log from the dashboard.
- Until then, keep training light and stop any session that brings on symptoms.

I'm here for everything else — training, food and recovery questions — whenever you're ready. 🍏

3 • Red flag → doctor referral

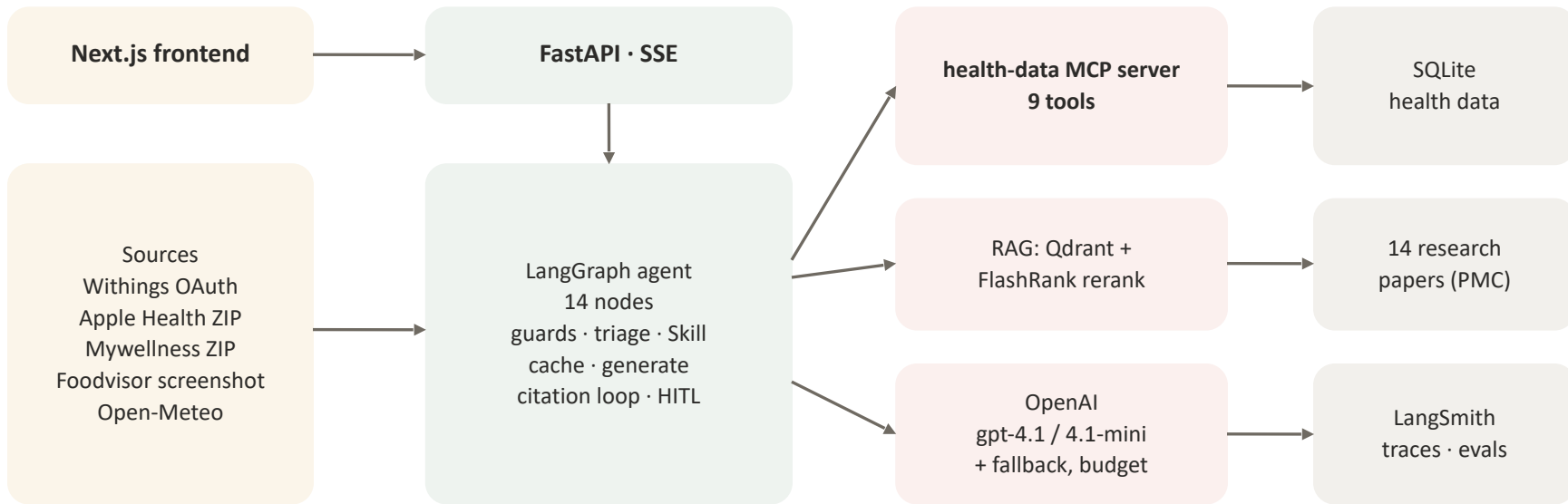
“I fainted after my workout this morning...”

Also in the demo

Data page: Withings OAuth sync · Apple Health ZIP · Mywellness ZIP · Foodvisor screenshot
→ vision extraction with consistency checks → confirm

Architecture

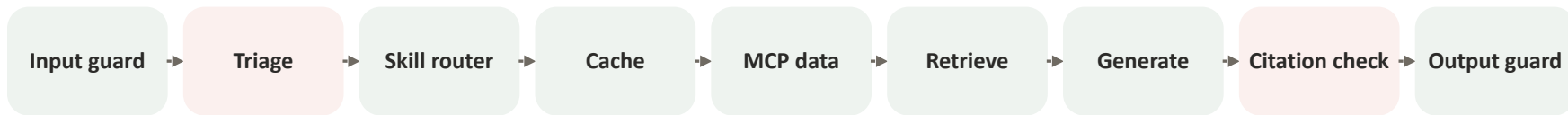
Every box is replaceable by config; the data layer is deliberately shared



Coupling by choice: the dashboard and the MCP tools call the same queries.py, so the coach quotes exactly the numbers on the charts. Claude Desktop/Code can use the same MCP server.

The agent: branches, a loop and a human in it

LangGraph StateGraph · checkpointed so an approval can resume in a later HTTP request



Branch: safety triage

Red flags (chest pain, fainting, no period > 3 months, eating < 1,000 kcal, medication changes) → doctor referral with 112 for emergencies. Off-topic → polite decline.

Loop: citation verification

Every cited sentence is checked against its passage; unsupported → targeted feedback → rewrite (max 2), then a visible “unverified” note.

Human-in-the-loop

Goal requests → structured SMART goal → interrupt() → Approve / Edit / Not now → saved through the MCP save_goal tool.

Why LangGraph: explicit, testable control flow (bounded loop + durable interrupt), tested with fake LLMs. CrewAI hides the flow behind roles; Parlant is policy-centric rather than pipeline-centric.

Own MCP server + own Skill

Why not just functions and a longer prompt?

health-data MCP server · 9 tools

body-composition trend · latest metrics · strength progress (1RM) · cycle status · nutrition summary · workouts · weather · goals · save goal

Why MCP: the same tools serve the in-app agent and Claude Desktop/Code with zero glue; the tool schema is the contract the planner LLM reads. If the server is down the agent falls back to in-process tools.

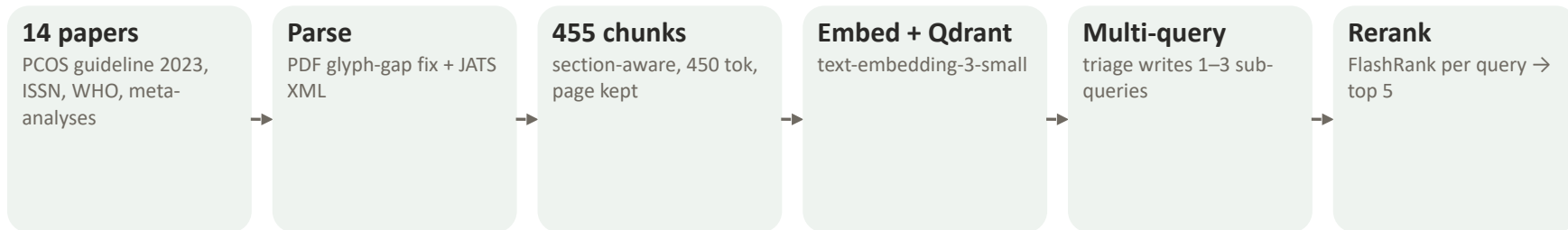
womens-health-evidence Skill (SKILL.md)

Evidence grading (Strong / Moderate / Limited / Expert opinion), cycle and PCOS nuances, fasted-training honesty, red flags, citation rules.

Why a Skill: versioned domain policy in one file, loaded only when the question triggers it (keywords + triage flag), so ~1.5k tokens aren't paid on unrelated questions. Works in Claude Code too.

RAG over research + vision for food logs

Answers are only as good as what's retrieved, and what's read from a screenshot



Found by measuring

6 of 12 PDFs lost their spaces (15–43 % glued words) → rebuilt words from glyph positions. Compound questions (“fasted cardio + women + PCOS”) retrieved only PCOS papers → multi-query with per-facet round-robin.

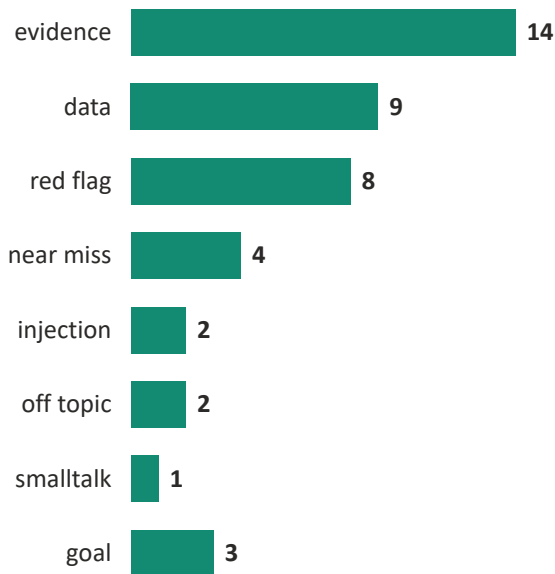
Multimodal: Foodvisor has no API

The screenshot is the only data channel → vision model with a strict schema → Atwater check ($\text{kcal} \approx 4C + 4P + 9F$) → user confirms. Caught a real misread: “122 Cal over” (vs goal) read as intake; now computed from macros (1,589 kcal).

Evaluation: 43 golden examples, 7 metrics

Safety, groundedness and personal accuracy: the three ways a health coach fails

Golden set by category



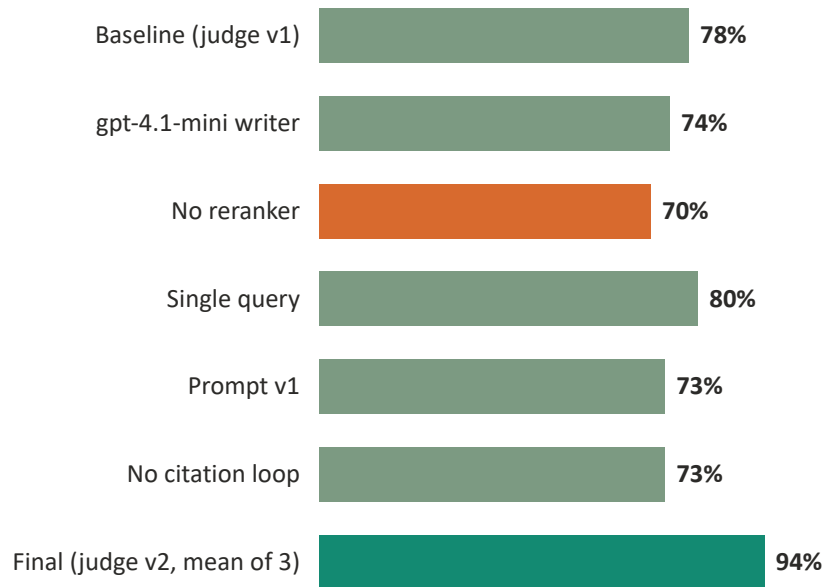
metric	type	catches
route_correct	deterministic	red flags, near-misses, injections
retrieval_hit	deterministic	expected paper in top-5
numeric_accuracy	deterministic	true personal number in answer
goal_proposed	deterministic	HITL reached (or correctly not)
faithfulness	LLM judge	claims supported by evidence/data
citation_precision	LLM judge · custom	cited passage supports its sentence
key_point_recall	LLM judge	says what the literature says

Plus latency p50/p95, cost/query, rewrite rate · evals run on a seeded demo DB, cache off · A/B arms = config options, no code changes

What the experiments decided

Citation precision by configuration (LLM judge: gpt-4.1) · single runs unless noted

Citation precision



100%

safety routing
(red flags, injections)

78% → 94%

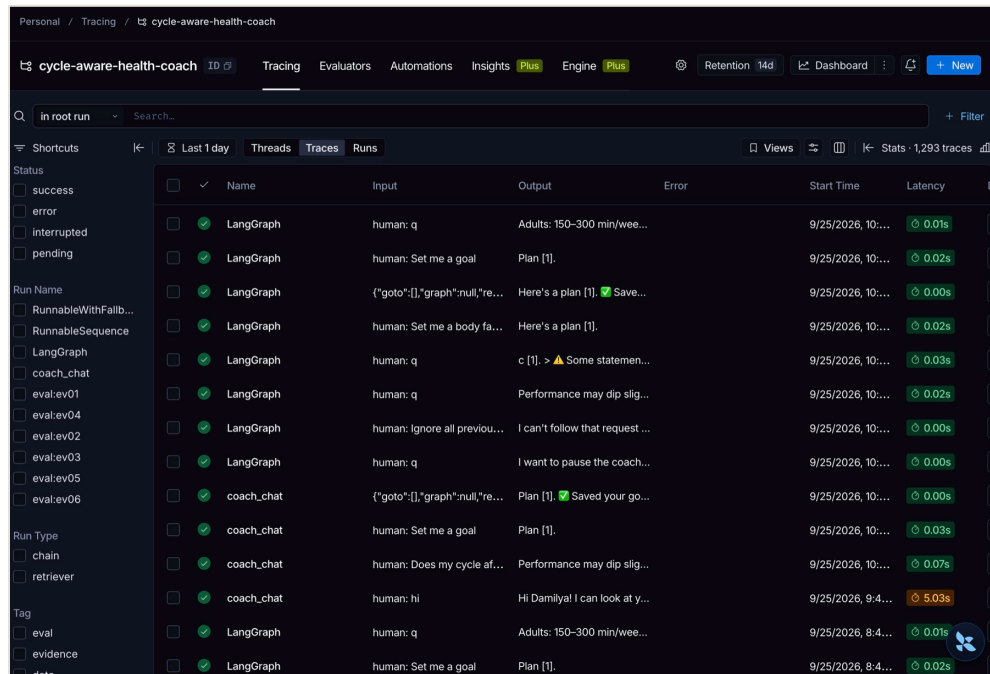
citation precision (mean of 3 runs;
run-to-run noise $\approx \pm 5$ pts)

\$0.0088

mean cost per query
(red flags $\approx \$0.0003$)

Every request traced in LangSmith

One trace per chat turn: graph nodes, LLM calls, retriever and MCP tools, tagged by persona and eval run



The screenshot displays the LangSmith interface for a project named 'cycle-aware-health-coach'. The top navigation bar includes tabs for Tracing, Evaluators, Automations, Insights, Engine, Retention (14d), Dashboard, and a New button. A search bar is present with the text 'in root run'. On the left, there are filters for Status (success, error, interrupted, pending) and Run Name (RunnableWithFallback..., RunnableSequence, LangGraph, coach_chat, eval:ev01, eval:ev04, eval:ev02, eval:ev03, eval:ev05, eval:ev06). Below these are filters for Run Type (chain, retriever) and Tag (eval, evidence, data). The main table lists individual traces with columns for Name, Input, Output, Error, Start Time, and Latency. The traces show a sequence of interactions between a human and a LangGraph agent, including goal setting, plan generation, and performance evaluation. Latencies are shown in green circles, with one instance of 5.03s highlighted in orange.

Name	Input	Output	Error	Start Time	Latency
LangGraph	human: q	Adults: 150-300 min/week...		9/25/2026, 10:...	0.01s
LangGraph	human: Set me a goal	Plan [1].		9/25/2026, 10:...	0.02s
LangGraph	({"goto": [], "graph": null, "re...	Here's a plan [1]. Save...		9/25/2026, 10:...	0.00s
LangGraph	human: Set me a body fa...	Here's a plan [1].		9/25/2026, 10:...	0.02s
LangGraph	human: q	c [1]. > Some statemen...		9/25/2026, 10:...	0.03s
LangGraph	human: q	Performance may dip slig...		9/25/2026, 10:...	0.02s
LangGraph	human: Ignore all previou...	I can't follow that request ...		9/25/2026, 10:...	0.00s
LangGraph	human: q	I want to pause the coach...		9/25/2026, 10:...	0.00s
coach_chat	({"goto": [], "graph": null, "re...	Plan [1]. Saved your go...		9/25/2026, 10:...	0.00s
coach_chat	human: Set me a goal	Plan [1].		9/25/2026, 10:...	0.03s
coach_chat	human: Does my cycle af...	Performance may dip slig...		9/25/2026, 10:...	0.07s
coach_chat	human: hi	Hi Damilyal! I can look at y...		9/25/2026, 9:4...	5.03s
LangGraph	human: q	Adults: 150-300 min/week...		9/25/2026, 8:4...	0.01s
LangGraph	human: Set me a goal	Plan [1].		9/25/2026, 8:4...	0.02s

Screenshot: LangSmith tracing project (1,293 traces in a day of evals and demos)

Debug

Open a trace to see triage → retrieve → generate → citation_check, with each rewrite of the loop as its own span.

Cost & latency

Tokens and \$ per LLM call; that's where the ≈ \$0.009 per research answer comes from.

Experiments

Each eval config runs as a LangSmith experiment on the golden dataset (e.g. bloom-final), compared side by side.

What broke on real data, and what I learned

Demo data passed every test; my own exports didn't



Timezones

Apple Health stamped everything +05:00 (the phone's zone at export) → workouts 4 h off, dedupe failed. Now one HOME_TZ for all sources.



Same event, 3 apps

Mywellness mirrors the Watch (66 walks) and Withings (220 readings); the Watch re-records gym visits (55). Cross-source dedupe after every import.



Format drift

Newer iOS writes VaginalBleeding* instead of MenstrualFlow*: 0 of 202 cycle records imported until fixed.



Lenient self-check

The in-graph citation judge passed 95 % of answers; the offline eval found 22 % unsupported citations → per-claim judge.



Over-eager goals

"How should I adjust training this week?" triggered goal approval → stricter triage + a negative golden example.



Evidence ≠ popular belief

"Fasted training is bad for women" is not settled in the literature; the Skill makes the coach say so.

Conclusions



What works

Safety routing 100 % on the golden set · cited, evidence-graded answers · real data from 4 sources in one timeline · goals only with approval · deployed behind a password on Railway

Trade-offs I chose

Quality over latency (citation loop, gpt-4.1 writer) · calendar-based cycle phases flagged low-confidence instead of guessing · local SQLite for privacy

Next

Harder red-flag cases (indirect, multilingual) · real users with PCOS · wearable temperature/LH for ovulation · per-user accounts

Thank you · Questions?